

Deep Learning Unit 1 - Quick Notes

Introduction to Artificial Intelligence (AI)

AI is the science of creating systems that can perceive, reason, learn and make decisions. Applications include healthcare, finance, robotics, cybersecurity, autonomous vehicles and virtual assistants. AI evolved from rule-based systems to modern machine learning and deep learning.

Machine Learning

Machine Learning enables systems to learn patterns from data. Types: Supervised Learning (labeled data), Unsupervised Learning (unlabeled data) and Reinforcement Learning (reward-based learning). AI is the broad field, ML is a subset of AI and Deep Learning is a subset of ML.

Introduction to Deep Learning

Deep Learning uses multi-layer neural networks for automatic feature learning. It is widely used in image recognition, NLP, speech recognition and recommendation systems. Advantages include high accuracy and automatic feature extraction; limitations include large data requirements and computational cost.

Biological Neuron

A biological neuron consists of dendrites, cell body, axon and synapses. Artificial neurons are inspired by this information-processing model.

Artificial Neural Networks

An artificial neuron multiplies inputs by weights, adds a bias and applies an activation function. Networks of neurons learn complex patterns.

Perceptron

A perceptron is the simplest neural network used for linear classification. It updates weights using prediction error.

Multilayer Perceptron

MLP contains input, hidden and output layers. Hidden layers enable learning of nonlinear relationships.

Activation Functions

Binary Step: binary output. Linear: identity. Sigmoid: 0-1 output. Tanh: -1 to 1. ReLU: $\max(0, x)$. Leaky ReLU: avoids dead neurons. Softmax: probability distribution for multi-class classification.

Feed Forward Neural Networks

Data flows from input to output only. Forward propagation computes predictions layer by layer.

Backpropagation

Computes gradients using the chain rule and updates weights with gradient descent to minimize loss.

Training

Key concepts: epoch, batch, mini-batch, learning rate and loss function.

Hyperparameters

Learning rate, epochs, batch size, hidden layers and neurons significantly affect model performance.

Challenges

Overfitting, underfitting, vanishing gradients and exploding gradients are common deep learning issues.

Optimization

Gradient Descent, Stochastic Gradient Descent and Mini-batch Gradient Descent are widely used optimization methods.

Applications

Computer vision, speech recognition, NLP, recommendation systems, autonomous driving, fraud detection and medical diagnosis.